

GNR 638 - Assignment 2 Computational Budget Report

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1 Hardware and Environment Specifications

All model training, feature extraction, and evaluation processes were executed utilizing cloud-based GPU infrastructure. The primary accelerator utilized for all experiments was a single NVIDIA GPU. The environment was configured using PyTorch with standard mixed-precision training enabled to optimize tensor core utilization.

2 Architectural Complexity Metrics

Before calculating the time-based computational budget, it is necessary to benchmark the inherent structural complexity of each tested architecture. Table 1 outlines the raw parameter counts, Multiply-Accumulate Operations (MACs), and Floating Point Operations (FLOPs) required for a single forward pass of an image.

Model	Total Parameters (M)	MACs (G)	FLOPs (G)
ResNet50	23.570	4.132	8.264
EfficientNet-B0	4.004	0.384	0.769
ConvNeXt-Tiny	27.820	4.455	8.910

Table 1: Model Complexity Profiling Metrics.

The metrics highlight a vast discrepancy in efficiency. EfficientNet-B0 operates as the most lightweight architecture by a significant margin, requiring over 10x fewer FLOPs than ConvNeXt-Tiny.

3 Training Runtime Analysis

The differences in FLOPs and parameter counts directly translate into observable differences in wall-clock training time. Table 2 details the average time required to complete a single epoch of training (forward pass, backward pass, and optimizer step) during our fine-tuning experiments over the Aerial Images Dataset (AID).

Model	Average Time per Epoch (s)	Estimated Time for 30 Epochs (min)
ResNet50	~ 34.7 seconds	~ 17.3 minutes
EfficientNet-B0	~ 27.6 seconds	~ 13.8 minutes
ConvNeXt-Tiny	~ 108.2 seconds	~ 54.1 minutes

Table 2: Runtime metrics observed during standard fine-tuning tasks.

4 Summary of Computational Expenditure

Over the course of five rigorous experimental scenarios (Linear Probing, Multiple Fine-Tuning Strategies, Few-Shot splits, Corruption Robustness tests, and Layer-Wise Feature Extraction), the models were trained and evaluated iteratively.

- **Most Efficient:** EfficientNet-B0 utilized the lowest computational budget. Its constrained design allowed for rapid experimentation, averaging under 28 seconds per epoch.
- **Most Expensive:** ConvNeXt-Tiny consumed the vast majority of our computational budget. Despite having a parameter count similar to ResNet50, the architectural overhead of emulating Vision Transformer patchification and large 7×7 convolutions resulted in an average training time of ~ 108 seconds per epoch—nearly four times slower than EfficientNet-B0.

Ultimately, EfficientNet-B0 proved to be the optimal choice not only for predictive accuracy but for strict computational budgeting in resource-constrained environments.